Network Device Discovery and Profiling

Local network assessment using Nmap on Kali Linux (VirtualBox bridged adapter)

Scope	Tools	Privacy
Discovery of active hosts, then focused analysis of three selected devices.	Kali Linux, Nmap, VirtualBox bridged adapter.	Internal IPs and identifying data were anonymized before publication.

Executive Summary

This report documents a small-scale local network assessment performed from Kali Linux running in VirtualBox with the network adapter set to bridged mode. The workflow started with host discovery across the 192.168.1.0/24 subnet, followed by deeper analysis of three representative devices: an end-user device, a smart/IoT-style device, and a Huawei network infrastructure device.

The results show three different exposure profiles. Device 1 had no exposed TCP services and presented a low external attack surface. Device 2 exposed UDP-based local discovery behavior consistent with multicast DNS (mDNS), which increases visibility on the LAN. Device 3 exposed HTTP, HTTPS, and DNS services and appears to be a Huawei ONT/router, making it the most critical asset in the observed environment.

Methodology

1. Performed host discovery with nmap -sn 192.168.1.0/24 to identify active devices.
2. Selected three devices with different behavior profiles for detailed analysis.
3. Used targeted Nmap scans (-A, -sU, and focused UDP scans) to profile services and host behavior.
4. Interpreted findings conservatively, using only evidence visible in scan output.

Findings Overview

Device	Role	Observed Exposure	Key Evidence	Risk View
Device 1	End-user device	No open TCP ports	1000 closed TCP ports; no exact OS match	Low
Device 2	Smart / IoT device	UDP 5353 visible	5353/udp zeroconf; additional UDP probing behavior	Moderate
Device 3	Huawei gateway / ONT	53/tcp, 80/tcp, 443/tcp open	Huawei certificate; DNS and web management services	High

1. Network Discovery

```
┌──(kali㉿Kali)-[~]
└─$ nmap -sn 192.168.1.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-18 11:55 UTC
Nmap scan report for h[redacted] (192.168.1.[redacted])
Host is up (0.057s latency).
Nmap scan report for 192.168.1.[redacted]
Host is up (0.033s latency).
Nmap scan report [redacted] (192.168.1.[redacted])
Host is up (0.050s latency).
Nmap scan report for 192.168.1.[redacted]
Host is up (0.015s latency).
Nmap scan report for 192.168.1.[redacted]
Host is up (0.13s latency).
Nmap scan report f[redacted] (192.168.[redacted])
Host is up (0.19s latency).
Nmap scan report [redacted] (192.168.1[redacted])
Host is up (0.077s latency).
Nmap scan report fo[redacted] (192.168.1.[redacted])
Host is up (0.0010s latency).
Nmap scan report for [redacted] (192.168.1.[redacted])
Host is up (0.12s latency).
Nmap scan report for 192.168.1.[redacted]
Host is up (0.079s latency).
Nmap scan report for 192.168.1.[redacted]
Host is up (0.12s latency).
Nmap done: 256 IP addresses ([redacted] hosts up) scanned in 8.89 seconds
```

Figure 1. Initial host discovery across the local subnet. The scan identified multiple active hosts and created the shortlist for deeper analysis.

Host discovery was performed first to establish the visible device population on the local network. This step provided the starting point for selecting devices with different exposure patterns for focused assessment.

2. Device 1 - End-User Device Profile

```
┌──(kali㉿Kali)-[~]
└─$ sudo nmap -A 192.168.1.[redacted]
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-18 12:06 UTC
Nmap scan report for A[redacted] (192.168.1.24)
Host is up (0.019s latency).
All 1000 scanned ports on [redacted] (192.168.1.[redacted]) are in ignore
d states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 3A:[redacted] (Unknown)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   19.10 ms [redacted] (192.168.1.[redacted])

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.92 seconds
```

Figure 2. Detailed scan of Device 1 showing no open TCP services and a largely closed attack surface.

Device 1 responded to host discovery and advanced scanning, but all 1000 scanned TCP ports were closed. Nmap was unable to derive a reliable operating system fingerprint because there were no exposed services to support stronger identification.

Interpretation: this is consistent with a typical user device that initiates outbound traffic but does not present administrative or application services to the rest of the LAN. From a remote exposure perspective, it is the lowest-risk device in the sample.

3. Device 2 - Smart / IoT Device Profile

```
┌──(kali㉿Kali)-[~]
└─$ sudo nmap -sU 192.168.1.
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-18 12:19 UTC
Nmap scan report for 192.168.1
Host is up (0.0052s latency).
Not shown: 966 closed udp ports (port-unreach), 33 open|filtered udp ports (n
o-response)
PORT     STATE SERVICE
5353/udp open  zeroc
MAC Address: B0                (Hon                .)

Nmap done: 1 IP address (1 host up) scanned in 7.46 seconds

┌──(kali㉿Kali)-[~]
└─$ sudo nmap -sU -p 5353 -Pn 192.168.1.
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-18 12:28 UTC
Nmap scan report for 192.168.1
Host is up (0.040s latency).

PORT     STATE SERVICE
5353/udp open  zeroconf
MAC Address: B0          :          (Hon          is  n          )

Nmap done: 1 IP address (1 host up) scanned in 0.60 seconds
```

Figure 3. UDP-focused scans on Device 2 showing zeroconf / mDNS visibility on port 5353.

Device 2 was more interesting from a LAN exposure perspective. A UDP scan identified 5353/udp (zeroconf / mDNS), which is commonly associated with service advertisement and local discovery workflows on smart devices, multimedia devices, or embedded systems.

Interpretation: this behavior suggests the device participates actively in local network discovery. That does not by itself indicate compromise, but it does increase the device's visibility and expands the number of network interactions that should be reviewed in a security assessment.

4. Device 3 - Huawei Network Infrastructure Device

```
┌──(kali㉿Kali)-[~]
└─$ sudo nmap -A 192.168.1
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-18 12:35 UTC
Nmap scan report for 192.168.1
Host is up (0.029s latency).
Not shown: 993 closed tcp ports (reset)
PORT     STATE    SERVICE   VERSION
/tcp   filtered ftp
/tcp   filtered ssh
53/tcp   open     domain    (generic dns response: REFUSED)
80/tcp   open     http
|_http-title: Site doesn't have a title (text/html; charset=UTF-8).
| fingerprint-strings:
|   FourOhFourRequest, GetRequest:
|     HTTP/1.1 400 Bad Request
|     Connection: Keep-Alive
|   GenericLines:
|     HTTP/1.1 404 Not Found
|     Connection: Keep-Alive
|     requested URL was not found on this server.
|   HTTPOptions, RTSPRequest, SIPOptions:
|     HTTP/1.1 404 Not Found
|     Content-Type:text/html; charset=UTF-8
|     Pragma:no-cache
|     Cache-control:no-cache, no-store, max-age=0
|     Transfer-Encoding:chunked
|     X-Frame-Options:SAMEORIGIN
|     Connection:Keep-Alive
|     X-XSS-Protection:1; mode=block
|     Content-Security-Policy:default-src 'self' 'unsafe-inline' 'unsafe-eval'
|_    requested URL was not found on this server.
443/tcp  open     ssl/https
|_ssl-date: 1981-02-12T14:3  :4  +0  :  ; -45y  d21  56m44s from scanner time.
| ssl-cert: Subject: commonName=ont.huawei.com/organizationName=Huawei/countryName=CN
| Not valid before: 2020-08-25T06:59:49
```

Figure 4. Advanced scan of the Huawei gateway / ONT showing exposed DNS, HTTP, and HTTPS services.

Device 3 is the most security-relevant host in the observed set. The scan exposed TCP port 53 (DNS), 80 (HTTP), and 443 (HTTPS), and the HTTPS certificate referenced ont.huawei.com, strongly indicating a Huawei gateway / ONT or similar network edge device.

Interpretation: unlike the other sampled devices, this host provides management and infrastructure-facing services. Because it sits in a critical network role, compromise of this device would have a disproportionate impact compared with an ordinary endpoint.

5. Comparative Analysis

The three selected devices demonstrate why device context matters in network analysis. A closed endpoint can be present and functional while still presenting minimal attack surface. A smart device may expose discovery-oriented traffic without looking overtly dangerous. A router or ONT, however, is both visible and structurally important, making it the highest-priority asset for hardening and review.

6. Security Recommendations

Priority Asset	Recommended Action	Rationale
Huawei gateway / ONT	Restrict management access and review firmware / admin exposure.	This device exposes web and DNS services and represents the highest-impact target.
Smart / IoT device	Review service discovery settings and segment IoT where possible.	mDNS-style behavior increases LAN visibility and can broaden attack surface.
End-user device	Maintain host firewalling and patch hygiene.	No exposed services were observed, which is good posture worth preserving.

7. Limitations and Privacy Note

This report is intentionally limited to the evidence captured during the performed scans. It does not claim full operating system identification where fingerprints were ambiguous, and it does not infer compromise without direct indicators. Internal addresses and other identifiers were masked for safe publication.

8. Conclusion

The assessment successfully moved from broad network discovery to host-by-host profiling. Among the three selected devices, the Huawei gateway / ONT is the most important security asset due to exposed infrastructure services, Device 2 presents moderate visibility through discovery-oriented UDP behavior, and Device 1 shows the lowest observed exposure. This progression - discover, classify, prioritize - is the core value of the project.